

ISAAC RIDER

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EDUCATION

University of Missouri

August 2022 – May 2026

Bachelor of Science in Computer Science / Minor in Mathematics

Columbia, Missouri

EXPERIENCE

Cofounder & Software Engineer, Tonetta

December 2025 – July 2026

Tonetta – Development Team

Columbia, Missouri

- Architected the Tonetta desktop application using the **Electron framework**, building a **modular frontend** with cross-platform support now running in production for **25+ users** on Windows and macOS.
- Engineered a **scalable Python-based backend** for **AI-driven feature extraction**, optimizing data pipelines to sustain a **sub-150ms real-time feedback loop** between the desktop client and AI models.
- Led **3 engineering interns** to implement complex **audio processing features**, including **WPM deceleration** and **real-time voice calibration** modules within the Electron environment.

Applications Development Intern

May 2025 – March 2026

University of Missouri – Applications Development Team

Columbia, Missouri

- Developed and maintained **four enterprise web applications** using **Java, Vaadin Flow 24, and Oracle SQL** within a high-availability distributed service architecture, used daily by **UM System administrators**.
- Applied **Object-Oriented Programming (OOP)** principles to build **modular, reusable UI components** and backend services, improving maintainability across the shared codebase.
- Collaborated in **Agile/Scrum** sprints, using **Git, Linux workflows, and SQL stored procedures** to deliver production-ready features alongside senior developers.
- TED (Today's Employee Database)**: Engineered a centralized system for Mizzou administration to manage and audit mandatory employee training records across the UM System.

Undergraduate Research Assistant

June 2024 – May 2026

Department of Electrical Engineering and Computer Science

Columbia, Missouri

- Built **computational models** of bladder function in **Python**, integrating neurophysiological data and validating model accuracy against experimental datasets.
- Applied software engineering practices—**Git**, modular design, and iterative testing—to build reliable scientific computing tools.

PROJECTS

PipeDreams Capital | *Python, React, SQL, REST APIs, Machine Learning*

Senior Capstone

- Built a full-stack **AI-powered stock trading platform** with portfolio management, real-time price tracking, transaction execution, and customizable price alerts.
- Designed a **normalized relational database** modeling users, portfolios, trades, and ML predictions as the core data layer.
- Implemented **machine learning models** (ARIMA, CNN, LSTM) generating buy/sell/hold signals with confidence scores, and integrated the **Alpaca API** to place autonomous trades.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, SQL, JavaScript, HTML, CSS

Frameworks/Tools: React, Vaadin Flow 24, Electron, Node.js, Git, REST APIs, Oracle DB, MySQL, VS Code

Platforms/Systems: Linux, Windows, macOS

Concepts: Object-Oriented Programming, Agile/Scrum, distributed systems, data modeling, machine learning, unit testing

LEADERSHIP

Boy Scouts of America

August 2018 – July 2025

Assistant Scoutmaster/Eagle Scout

Independence, Missouri

- Earned the rank of **Eagle Scout**; mentored younger scouts through rank advancement and led outdoor activities teaching survival, navigation, and **teamwork**.